PS#6: Density and center of mass - Answer key quick ref

1. (APC 6.3 Exercise 6)

   (a) Find the mass of each bar.

   The mass of bar ρ is 1.4711 grams.

   The mass of bar p is 6.3212 grams.

   (b) Find the center of mass of each bar.

   The center of mass of bar ρ is 1.5686 cm.

   The center of mass of bar p is 4.1802 cm.

   (c) Now consider a new 10 cm bar whose mass density function is $f(x) = \rho(x) + p(x)$.

   i. Explain how you can easily find the mass of this new bar with little to no additional work.

   The mass of the new bar is 7.7923 grams.

   ii. Similarly, compute $\int_0^{10} xf(x)\,dx$ as simply as possible, in light of earlier computations.

   The sum of the two previous numerators: 28.7317 g $\cdot$ cm.

   iii. (New CoM)

   $\frac{28.7317}{7.7923} \approx 3.6872$ cm

2. (AC 6.3 Exercise 7) Consider the curve given by $y = f(x) = 2xe^{-1.25x} + (30-x)e^{-0.25(30-x)}$.

   (a) (Plot)

   (b) Let x and y be measured in inches. Find the total volume of the baseball bat generated by revolving the given curve about the x-axis. Include units on your answer.

   We should certainly slice vertically, and then our rectangles will revolve into disks. Therefore, the volume is given by

   $$V = \int_{x=0}^{x=30} \pi[f(x)]^2\,dx = \int_{x=0}^{x=30} \pi[2xe^{-1.25x} + (30-x)e^{-0.25(30-x)}]^2\,dx \approx 52.1 \text{ in}^3$$

   (Thanks, Wolfram|Alpha!)

   (c) Suppose that the baseball bat has constant weight density, and that the weight density is 0.6 ounces per cubic inch. Find the total weight of the bat whose volume you found in (b).

   31.26 oz.

   (d) (Explain setup)

   (e) Compute the center of mass of the baseball bat.

   Center of mass: $\frac{725.21 \text{ oz} \cdot \text{in}}{31.26 \text{ oz}} \approx 23.2$ in.